

No.	Solution
1(a)	The resultant force acting on a body is zero. The resultant torque acting on the body about any pivot point is zero.
1(b)	Resolving forces vertically, $T_1 \sin(20^\circ) + T_2 \sin(10^\circ) = 700$ --- (1) Resolving forces horizontally, $T_1 \cos(20^\circ) = T_2 \cos(10^\circ)$ $T_2 = T_1 \frac{\cos(20^\circ)}{\cos(10^\circ)} \text{ --- (2)}$ Substitute (2) into (1), $T_1 \left[\sin(20^\circ) + \sin(10^\circ) \left(\frac{\cos(20^\circ)}{\cos(10^\circ)} \right) \right] = 700$ $T_1 = 1378.7$ $\approx 1400 \text{ N}$ $T_2 = 1378.7 \times \frac{\cos(20^\circ)}{\cos(10^\circ)}$ $= 1315.5$ $\approx 1300 \text{ N}$
1(c)	Angle of support cable above horizontal = θ $\tan \theta = \frac{1.2}{1.6} = 0.75$ $\theta = \tan^{-1}(0.75)$ $= 36.87^\circ$ Taking moments about the base of pole, $150 \cos(10^\circ) \times 1.8 = T \cos \theta \times 1.2$ $T = \frac{150 \cos(10^\circ) \times 1.8}{\cos(36.87^\circ) \times 1.2}$ $= 276.98$ $\approx 280 \text{ N}$
2(a)(i)	A transverse wave is a wave in which the oscillation of the particles is perpendicular to the propagation of the wave energy.